Negative Results for Elliptic-Curve Index Calculus: Six Statements, the Proofs or Evidence Behind Them, and What They Do Not Say

Anonymous

Abstract

We collect six negative results on index calculus for the elliptic-curve discrete logarithm and state each at the tier its evidence supports. (1) The compact-orbit Koblitz decomposition configurations measured in this programme sit on or above the generic preprocessing line $ST^2 \gtrsim r/|\mathrm{Aut}|$, and at the rank-stage mean on the non-adaptive line $ST \approx r/|\mathrm{Aut}|$. (2) Isogeny oracles add nothing in a generic group model: we reprove the bound $(Q+2)^2/r + 1/r$ and give a Lean statement of its counting core. (3) The mean decomposition yield of a factor base of size B is $\binom{B+m-1}{m}/N$ whatever its geometry. (4) The monodromy group of the Semaev summation cover is $(\mathbb{Z}/2)^{m-2}$; this is a derivation that has not had an independent review. (5) Lemma 4 of Nagao's ePrint 2015/984, $d_F \le d'_F$, is false over $\mathbb{F}_2$ under the equality reading of its definitions; one witness is checked by hand and both are re-derived natively, and nothing is claimed about the paper's Proposition 5. (6) We could not establish that Semaev's Assumption 1 fails off the diagonal, and state what is and is not known. Every number links to a committed evidence file. None of this is a speedup or a statement about a deployed curve.

1 Introduction

A programme that searches for faster index calculus on elliptic curves produces many negative results, and they are easy to lose: each sits in a different repository, at a different strength, behind a different convention. This paper states six of them in one place. For each we say what is proved, what is an exact finite computation, what is a measured committed record, and what is only a derivation that nobody outside the programme has read. We also say what each statement does *not* say.

The six statements are:

S1 compact-orbit Koblitz index calculus is not evidence of a non-generic speedup: its measured configurations sit on or above the generic preprocessing lines (Section 2);

S2 isogeny oracles add nothing in the generic group model (Section 3);

S3 the mean decomposition yield is conserved under factor-base geometry (Section 4);

S4 the monodromy of the Semaev summation cover is $(\mathbb{Z}/2)^{m-2}$ (Section 5);

S5 Lemma 4 of Nagao's ePrint 2015/984 is false as a universal statement (Section 6);

S6 whether Semaev's 2015 Assumption 1 fails off the diagonal is *not* established by the programme's records (Section 7).

S6 is included because the design of this paper listed it as a negative result, and the evidence on disk does not support that. Reporting it as open is the result.

Claim class. Under the programme's success criterion a relabelling or an accounting change is not an improvement, and this paper is that: a consolidation of existing negatives. It measures no new speed, reports no gap to Pollard rho of its own (the studies it cites report theirs), and makes no key-recovery, sub-rho or deployed-curve claim.

Labels. Each statement carries one label, fixed by its evidence and not by our judgement of how convincing it is. PROVED: a proof is in the paper. EXACT: an exact finite computation that we re-ran natively, with its scope stated. RECORDED: committed measured rows, cited by file and digest. DERIVED, UNREVIEWED: a written derivation that has not had an independent review. NOT ESTABLISHED: the records do not support the statement. Table 1 lists them.

	statement	label	evidence
S1	compact-orbit IC on or above $ST^2 \gtrsim r/\lvert\mathrm{Aut}\rvert$	RECORDED (14 rows)	[E1], [E2], [E3]
S2	isogeny oracles add nothing generic	PROVED; counting core in Lean (not yet kernel-checked)	[E4], [E31]
S3	mean yield conserved under base geometry	PROVED; EXACT; RECORDED	[E5], [E7], [E6], [E18], [E32]
S4	Semaev cover monodromy $(\mathbb{Z}/2)^{m-2}$	DERIVED, UNREVIEWED	[E9], [E10], [E11], [E13]
S5	Nagao Lemma 4 false (equality reading, $\mathbb{F}_2$)	EXACT	[E14], [E15], [E18]
S6	Assumption 1 fails off the diagonal	NOT ESTABLISHED	[E20], [E21], [E22]

Table 1: The six statements, their labels and evidence. Identifiers E_i are rows of the evidence ledger (Appendix A).

Conventions. Evidence files live in two repositories and are cited with the prefix `crypto:` or `crypto-autoresearcher:`; Appendix A gives each file's SHA-256. r is the prime order of the subgroup under attack, $A = |\mathrm{Aut}|$ is the order of the automorphism group the base is folded by, S is the stored advice (table entries, not bits), T is the online cost in the unit named in each section. All measurements are at toy scale. A *probe* (one quadratic solve plus a root-table lookup) is not a group operation; where a bound is stated in group operations the section says so.

2 S1: compact-orbit Koblitz index calculus against the generic lines

Setting. The compact-orbit configuration on a Koblitz curve over $\mathbb{F}_{2^n}$ folds a factor base by the Frobenius orbit, collects 4-sum relations through a table of roots of the third summation polynomial, and extracts a target by a fixed state-and-shift scan of that table. The table has $S = K^2 n$ entries. The study we rely on ([E3], [E1], [E2]) audits the committed records of this configuration for $n \in \{37, 41, 53, 61, 71, 73, 83\}$ against three published lower bounds, and we use only its *adaptive* and *non-adaptive* preprocessing lines:

- the adaptive line $S\,T^2 \approx r$ (Bernstein–Lange; Corrigan-Gibbs and Kogan, which also gives the matching algorithm) [1, 2];
- the non-adaptive line: a non-adaptive algorithm with advice S and T queries succeeds with probability at most $3T^2/r + 4\ln 2 \cdot ST/r$, so below $T = \sqrt{r}$ it needs $ST \gtrsim r$ [3].

Dividing r by A is the audit's own convention ([E3]: "Theorem 3.2 has none"), not a theorem of the cited papers. We keep that division and say so.

Result. Table 2 reproduces the audit's compact-orbit ladder. The first block uses the single frozen target probes; the second the rank-stage mean probes per relation.

Proposition 1 (S1, RECORDED)**.** *On the seven rungs of Table 2: (a) every measured row, whether the single frozen target or the rank stage plus target, has* $\log_2(ST^2) - \log_2(r/A) \geq 7.1$ *(frozen target:* 7.1 *to* 25.4*; rank stage plus target:* 16.8 *to* 45.8*); (b) at the rank-stage mean probes per relation,* $ST/(r/A)$ *lies in* [1.053, 2.734]*; (c) for single frozen targets* $ST/(r/A) < 1$ *at n37, n41, n53, n71 and n73.*

The audit also records, for the whole study, that none of 208 configurations (prime, random binary, Koblitz round five, one-target and batch Koblitz, and the compact-orbit probe rows) has a measured cost below its bound ([E3], Result 1; smallest measured-to-bound ratios 38.9, 56.8, 355, 2.84×10^6, 7.99×10^4 and 1077 for the six families).

rung	$\log_2 r$	A	$S = K^2 n$	$\log_2(ST^2) - \log_2(r/A)$ frozen target	$ST/(r/A)$ frozen target	$ST/(r/A)$ rank-stage mean
n37	27.78	74	1.813×10^3	9.4	0.627	1.094
n41	39.00	82	2.962×10^5	14.3	0.939	1.092
n53	44.26	106	2.565×10^6	13.1	0.342	1.081
n61	47.21	122	2.196×10^7	19.7	3.741	1.053
n71	52.29	142	2.556×10^7	7.1	0.010	1.478
n73	56.26	146	2.628×10^7	13.2	0.020	2.734
n83	52.93	166	2.988×10^7	25.4	5.120	1.417

Table 2: Compact-orbit ladder ([E2], rows `target_probes_online` and the rank-stage-mean sentence; same numbers in [E1]). T is in probes, not group operations.

How to read (c). The non-adaptive bound limits the *success probability* of an algorithm over random targets; a single cheap target is not a counterexample. The audit finds the two cheapest draws, n71 and n73, at 0.0065 and 0.0075 times the rank-stage mean probes per relation; under an exponential first-hit law each is a $P \approx 0.006$–0.007 event, and it calls them improbably cheap ([E3], "Record problems", item 3). We report them and do not average them away.

What S1 does not say. It is not a proof that the configuration is a generic algorithm. The structured generic-group bound is vacuous here: every rung has $\delta_{\text{upper}} = 1$ (its 4-sums cover the subgroup), so it certifies nothing ([E3], Result 1). The comparison is in probes and table entries, and the audit flags the rows `unit_not_group_operations`. The audit also found three record problems that we carry forward: the n61 subgroup order is recorded as $\log_2 r = 47.54$ in the ledger against the exact 47.2109; the excluded-precompute field means different things at different rungs; and the n71 and n73 frozen targets are improbable ([E3], "Record problems"). The distance to matched rho at full cost is outside this paper.

3 S2: isogeny oracles add nothing in the generic group model

Let $E_1, \dots, E_k$ be curves in one $\mathbb{F}_q$-isogeny class with $N = h\,r$, r prime and $r \nmid h$. Each order-r subgroup $G_j \subseteq E_j(\mathbb{F}_q)$ is encoded by an independent random injection $\sigma_j : \mathbb{Z}/r \to \{0,1\}^*$. The algorithm has oracles for addition and negation in every G_j and for every $\mathbb{F}_q$-rational isogeny $\psi : E_i \to E_j$ of any degree, restricted to G_i, and is given the scalars $\lambda_\psi \in \mathbb{Z}/r$ through which each ψ acts.

Theorem 2 (Theorem 15.1 of [E4]; PROVED)**.** *An algorithm that makes Q oracle calls finds $\log_{g_1}(Q_1)$ with probability at most $(Q+2)^2/r + 1/r$.*

Counting convention. The note does not say how the generators of the other G_j are supplied. We read the theorem as giving the algorithm the encodings of g_1 and Q_1 only, with every other element obtained through an oracle answer; then at most $Q+2$ distinct elements ever exist. If encodings of fixed generators of the other groups are also given, $Q+2$ becomes $Q+k+1$ below and nothing else changes.

Proof. Write $x = \log_{g_1} Q_1$, uniform in $\mathbb{Z}/r$. Because G_i has prime order, $\ker\psi \cap G_i$ is 0 or G_i, so a rational isogeny maps G_i either isomorphically onto G_j or to the identity (this includes isogenies of degree divisible by r whose kernel is the other Frobenius eigenline of $E[r]$). With the λ_ψ known, every element the algorithm can hold has the form $\sigma_j(a + bx)$ with $(a,b) \in (\mathbb{Z}/r)^2$ computed from the query structure and the λ_ψ alone: group operations add and negate forms, and an isogeny multiplies a form by λ_ψ or by 0.

The argument is Shoup's [7], adapted to forms on several curves. Game 0 is the real game. Game 1 is run by a simulator that tracks only the forms (a,b), giving the forms $(1,0)$ and $(0,1)$ to the inputs g_1 and Q_1. It answers each query with the string already assigned to the same form on the same curve, and with a fresh random string if the form is new. The simulator never uses x. The two games give the same transcript unless two *distinct* forms on one curve have the same value at x, the bad event. In Game 1 the algorithm's view is independent of x, so it outputs the right x with probability exactly $1/r$.

It remains to bound the bad event. Fix the simulator's random choices; they fix the families $P_1, \dots, P_k$ of forms issued on each curve, with $\sum_j |P_j| \le Q+2$. The set of secrets at which two different members of P_j coincide has at most $|P_j|^2 - |P_j|$ elements, by Lemma 3 below, which holds over any field. Summing

over j gives at most $(Q+2)^2$ bad secrets out of r. Averaging over the simulator's choices, the bad event has probability at most $(Q+2)^2/r$, so the real algorithm wins with probability at most $1/r+(Q+2)^2/r$. □

Lemma 3 (counting core)**.** *Let F be a field and $P \subseteq F \times F$ a finite set of affine forms (a, b), read as $x \mapsto a + bx$. (i) Two distinct forms agree at no more than one $x \in F$. (ii) For any finite $S \subseteq F$, the number of $x \in S$ at which two different members of P agree is at most $|P|^2 - |P|$. (iii) Off that set the evaluation map $p \mapsto a + bx$ is injective on P.*

Proof. (i) If $a+bx = c+dx$ and $a+by = c+dy$ with $(a, b) \neq (c, d)$: if $b = d$ then $a = c$, a contradiction; otherwise $(b-d)(x-y) = 0$ gives $x = y$. (ii) The bad set lies in the union, over the $|P|^2 - |P|$ ordered pairs of distinct forms, of the sets in (i). (iii) is the definition. □

The bound $|P|^2 - |P|$ counts ordered pairs; the unordered count $\binom{|P|}{2}$ holds as well, and is attained (see below).

Machine support. Lemma 3 is stated and proved in Lean in `IcNegative/AffineCore.lean` ([E31]; Section 9), and it is checked exhaustively on $\mathbb{Z}/p$ for $p = 5, 7, 11$ in the native instrument ([E18]): 2352 ordered pairs of distinct forms on $\mathbb{Z}/7$ and 14520 on $\mathbb{Z}/11$ agree at most once; over all 18424 sets of three forms on $\mathbb{Z}/7$ the largest bad set has $3 = \binom{3}{2}$ elements, and over all 211876 sets of four forms on $\mathbb{Z}/7$ it has $6 = \binom{4}{2}$. The negative control behaves as it must: on $\mathbb{Z}/8$ the forms $(0, 0)$ and $(0, 4)$ agree at 4 points, and on $\mathbb{Z}/9$ the forms $(0, 0)$ and $(0, 3)$ at 3, so the field hypothesis is needed.

What S2 does not say. Theorem 2 concerns isogeny *maps* used as oracles. Two things sit outside its hypotheses on purpose ([E4]): the Weil and Tate pairing on the $[r]$-fibre is not an isogeny oracle, and Smart's attack uses a lift to $\mathbb{Q}_p$. Any effect of an isogeny on index calculus must therefore come from the *representation*, the coordinates in which factor bases and summation polynomials live. The Lean statement is the counting step only; the oracle model, the simulator and the hybrid step are not formalised (`OPEN_TARGETS.md`, O1).

4 S3: mean decomposition yield is conserved under base geometry

Let G be a finite abelian group of order N and $D \subseteq G \setminus \{0\}$ a factor base of B distinct elements. For $m \geq 1$ and $g \in G$ let $c_D(g)$ be the number of size-m multisets from D whose sum is g.

Proposition 4 (yield conservation, PROVED)**.** $\sum_{g \in G} c_D(g) = \binom{B+m-1}{m}$. *Hence the mean per-target yield is $\binom{B+m-1}{m}/N$ for every base of size B, and the fraction of targets with at least one decomposition is at most* $\min\{1,\ \binom{B+m-1}{m}/N\}$.

Proof. Every size-m multiset has exactly one sum, so the multisets are partitioned by their sums; there are $\binom{B+m-1}{m}$ of them. The coverage bound is the pigeonhole bound on the image of a map from that set to G. □

The identity holds for any map from a finite domain, not only for multiset sums. Lean ([E32]): `sum_yield` is $\sum_g \mathrm{yield}(\varphi, g) = |\tau|$, `total_yield_eq` says two maps on domains of equal size have equal total yield, and `coverage_le_card_dom` and `coverage_le_card_cod` bound the number of covered targets by $\min(|\tau|, |G|)$. The list form of the same identity (`sum_count`, `patterns_neutral`) is already kernel-checked in [E7], with the axiom log [E8] showing only the standard foundations. The binomial count of the domain $\mathrm{Sym}^m(D)$ is not formalised (O2).

Corollary (screening rule). A proposal that promises a higher *mean* yield at matched base size is refuted by arithmetic before any experiment. Only coverage, multiplicity, relation rank, recognisability and the cost of finding a decomposition are free.

Exact and recorded support. (i) The programme's finding records that over 144 cells at $N \sim 2^{14}, 2^{16}, 2^{18}$ on 12 generated prime-order curves with $m = 3$ the largest absolute deviation of the exact cell mean from $\binom{B+2}{3}/N$ was exactly 0 ([E5], Consequence 1). (ii) The native instrument enumerates every base of size $B \in \{3, 4\}$ in $\mathbb{Z}/N$ for $N \in \{11, 12, 13\}$ (1540 bases in all) and finds the total always equal to $\binom{B+2}{3}$, while coverage varies, for example between 3 and 9 targets of 12 at $B = 3$, $N = 12$, and an arithmetic-progression base has the same total with coverage below the maximum ([E18]). (iii) The cyclotomic γ-weight base was tested against 1999 random Frobenius-stable bases of equal column count at $n = 23$: on all 10 coverage tests and all 5 energy tests it lies inside the null distribution, smallest two-sided $p = 0.363$ against a critical 0.00333, largest deviation from the null median 0.65%, while the positive control is detected at the floor $p = 0.001$ ([E6]). (iv) At toy sizes of 20 to 30 bits and $m = 2$, the matched-size yield ratios of the y-window arms against a random set of the same size lie inside $[0.95, 1.03]$ ([E33]).

What S3 does not say. It does not say that structured bases cannot beat random bases: coverage headroom up to $\min\{1, \mu\}$ is real (a Bose–Chowla Sidon base reaches it, [E5]), and under a one-relation-per-target solve the gain is at most $\min\{1, \mu\}/(1-e^{-\mu}) \leq 1.582$. It says nothing about the cost of finding a decomposition or the linear-algebra stage, where the cost of index calculus sits. The yield-neutral result of [E6] tests the yield clause only; the solving-degree clause of that lead was not tested there. In the campaign of 2026-10-18 the planned toy check of conservation inside the kernel-program search (item 10) has not run ([E23]); we cite no number from it. Item 17 ([E35]) measures the counted cost of a cyclotomic base at $n = 83$, not its yield: it reports that the base-dependent part of the cost is small next to the arm-independent pair table, and it is context for S3, not evidence for or against it.

5 S4: the monodromy of the Semaev cover is $(\mathbb{Z}/2)^{m-2}$

Let E be an elliptic curve over any field K, $\pi : E \to \mathbb{P}^1$ the quotient by inversion, $m \geq 3$ and $n = m - 1$. Let $t_1, \ldots, t_n$ be independent transcendentals, $L = K(t_1, \ldots, t_n)$, and P_i the tautological point with $\pi(P_i) = t_i$. The m-th Semaev summation polynomial $S_m(t_1, \ldots, t_n, T) \in L[T]$ has degree 2^{m-2} in T, and its roots are the x-coordinates of the signed sums $\epsilon_1 P_1 + \cdots + \epsilon_n P_n$.

Theorem 5 ([E9], [E10]; DERIVED, UNREVIEWED)**.** (A) *S_m is separable, its splitting field is the function field of E^n/Δ with Δ the diagonal inversion, and $\mathrm{Gal}(S_m/L) = (\mathbb{Z}/2)^n/\Delta \cong (\mathbb{Z}/2)^{m-2}$, acting simply transitively on the 2^{m-2} roots; arithmetic and geometric monodromy agree. The group is the same for every curve, base field and characteristic.* (B) *Over $\mathbb{F}_q$, at every specialisation outside an explicit degenerate locus of size $O_m(q^{m-2})$, $S_m(a, T)$ is squarefree and is either a product of 2^{m-2} distinct linear factors, when the n fibres of π over the a_i are simultaneously rational, or a product of 2^{m-3} irreducible quadratics; no factor of degree ≥ 3 and no mixed type occurs.* (C) *If every a_i is the x-coordinate of an $\mathbb{F}_q$-rational point, $S_m(a, T)$ splits completely, at every m: the factor-base locus lies inside the totally split locus.*

Derivation. The roots are the x-coordinates of the signed sums. The splitting field sits inside $K(E^n)$, which is Galois over $K((E/\pm)^n) = L$ with group $(\mathbb{Z}/2)^n$: it is the fibre product of n copies of a degree-2 Galois cover, so its deck group is $(\mathbb{Z}/2)^n$ by Artin's theorem, with no square-class computation and in every characteristic. The diagonal fixes every root since $x(R) = x(-R)$. Nothing else does: a further coincidence among the signed sums forces $\sum_{i \in S} \epsilon_i P_i \in E[2]$ for some proper nonempty S; the summation map $E^n \to E$ is a surjective homomorphism, so the preimage of the finite scheme $E[2]$ is a proper closed subscheme, and the generic point of the integral variety E^n lies in no proper closed subscheme. The group is abelian, so stabilisers are normal and the root field is the splitting field. The last step is the whole content: an earlier sketch of this result omitted it, and a correction records the gap and its repair ([E12]). The Frobenius element of a good specialisation is the class of the vector of fibre rationalities modulo global sign, and a non-identity element of a regular permutation group is fixed-point free, which gives (B). (C) holds because a rational point makes the decomposition group trivial.

Toy-scale verification. The verification run ([E11], seed 20260807) found on the factor-base locus that all 7908 good specialisations at $m = 3$, all 7038 at $m = 4$ and all 1161 at $m = 5$ split completely with roots equal to the signed sums, with no counterexample (8000, 8000 and 2000 samples, of which 92, 962 and 839 were degenerate). At $m = 3$ it saw 7786 good generic specialisations, exactly two Frobenius classes, only the patterns predicted, and maximum factor degree 2; the discriminant identity

$\mathrm{disc}_T S_3 = 16 f(x_1) f(x_2)$ holds exactly over $\mathbb{Z}[x_1, x_2, A, B]$. Two null objects of the same shape obey the dichotomy in only 13.69% and 48.91% of separable degree-4 samples, so the instrument can reject ([E11], `C7_null_controls`).

Status. The derivation is complete and self-contained but has not been read by a reader experienced in function fields; the two points that most need it are Artin's theorem for a quotient by a finite group acting faithfully, and properness of the preimage of $E[2]$ in characteristic 2 where $E[2]$ is non-reduced ([E34], "Two blockers"). A separately derived hypothesis reaches the same group, and its toy-scale execution received a validator and a red-team review ([E13]); that corroborates the statement and is not a proof review. Novelty has not been adjudicated.

What S4 does not say. It is not an algorithm. Statement (C) is about solving for the *last* coordinate given the other $m-1$; relation search asks the opposite question, with the target fixed and the other coordinates to be found in the factor base, and complete splitting gives it no leverage. Its negative content is that a census of factorisation types or Frobenius classes of the summation fibre is constant on the factor-base locus and so cannot detect a relation-rate anomaly there, and that no curve has smaller monodromy. For $m \geq 4$ the discriminant is a square in L, so a statistic built on its character is vacuous; $m = 3$ is the only m where it carries information. It implies no speedup, no security reduction and no change to any complexity estimate.

6 S5: Lemma 4 of Nagao's ePrint 2015/984 is false as a universal statement

Definition 6 (Definitions 5 and 6 of [5], equality reading)**.** Let $f_1, \ldots, f_M \in \mathbb{F}_2[X_1, \ldots, X_N]$. A *fall at degree* d is a tuple (g_i) with $\max_i \deg(g_i f_i) = d$, $\deg(\sum_i g_i f_i) < d$ and $\sum_i g_i f_i \neq 0$. The *first fall degree* d_F of a generating set is the least d at which a fall exists. The *fake* first fall degree d'_F of $\{f_i\} \cup S_{fe}$, with $S_{fe} = \{X_j^2 + X_j\}$, is defined the same way with every degree taken after reduction modulo S_{fe} and with $\sum_i g_i f_i \not\equiv 0$.

Lemma 4 of [5] asserts $d_F \leq d'_F$; the paper derives it from a lemma whose proof it declines to reproduce, and the fake quantity is what its applications bound ([E29]). The printed condition reads "$\max_i \deg(g_i f_i) \geq d$"; under that literal reading the lemma holds only vacuously ([E15]). We state the result for the equality reading, which is the one the complexity estimates need. We consider d_F with the field equations *outside* the generating set (F alone) and *inside* ($F \cup S_{fe}$).

Proposition 7 (S5, EXACT)**.** *Over $\mathbb{F}_2$ the statement $d_F \leq d'_F$ is false in both placements. Witness A: $f_1 = X^2Y$, $f_2 = XY + X$ in $\mathbb{F}_2[X, Y]$ has $d'_F = 2$ and $d_F = 3$ outside and inside. Witness B: $f_1 = X_1X_2 + X_3$, $f_2 = X_1X_3 + X_2$ in $\mathbb{F}_2[X_1, X_2, X_3]$ has $d'_F = 2$ and $d_F = 3$ outside and inside.*

Hand proof for Witness A. $d'_F = 2$. Reduced, $f_1 \equiv XY$, $f_2 = XY + X$. Take $g_1 = g_2 = 1$: both products have degree 2 and the sum is X, of degree $1 < 2$ and nonzero, a fall at 2. In the reduced ring with basis $1, X, Y, XY$, the multiples $g_1 \bar{f}_1$ lie in $\{0, XY\}$ and the multiples $g_2 f_2$ in $\{0, XY + X\}$, since $X \cdot f_2 = f_2$ and $Y \cdot f_2 = XY \cdot f_2 = 0$. A nonzero product has degree 2, so no tuple has maximum degree 1 and no fall occurs below 2. $d_F = 3$. At degree 3, $g_1 = 1$, $g_2 = X$ gives $f_1 + Xf_2 = X^2Y + X^2Y + X^2 = X^2$, with both products of degree 3 and the sum of degree 2 and nonzero: a fall. For $d \leq 2$, f_1 has degree 3 so $g_1 = 0$; f_2 has degree 2, so at $d = 2$ only g_2 constant, and cf_2 has degree 2. Adjoining $X^2 + X$ and $Y^2 + Y$ changes nothing: at $d = 2$ the leading forms XY, X^2, Y^2 of $f_2, X^2 + X, Y^2 + Y$ are linearly independent, so no nonzero combination has degree < 2, and no generator has degree ≤ 1. □

Native re-derivation. The instrument of [E19] computes d_F and d'_F exactly. For degree d the tuples with $\deg(g_i f_i) \leq d$ form an $\mathbb{F}_2$-vector space W_d; those whose degree-d part of the sum vanishes form a subspace K; $Z \subseteq K$ is the subspace with sum zero. A fall at d exists exactly when $K \not\subseteq Z$ and $K \not\subseteq W_{d-1}$, because a vector space is never a union of two proper subspaces. For d'_F the multipliers may be taken multilinear. The output ([E18]) is in Table 3.

The instrument also enumerates all 7875 unordered pairs of distinct nonzero multilinear polynomials of degree 1 or 2 in three variables with a cap of 6 on the degree. It finds $d_F > d'_F$ in 6138 pairs outside and 4830 inside, and $d_F < d'_F$ in none outside and 224 inside; no pair was undetermined at the cap. The population is an arbitrary finite one and the rates describe it only; they show that violations are

system	d_F outside	d_F inside	d'_F
control: X, $X+1$ (one variable)	1	1	1
Witness A: X^2Y, $XY+X$	3	3	2
Witness B: $X_1X_2+X_3$, $X_1X_3+X_2$	3	3	2
Witness C: $X_1X_2+X_3$, X_1+X_2	3	2	2

Table 3: First fall degrees from the native instrument ([E18]). Witness C is the ledger's first, outside-form example: it violates $d_F \le d'_F$ only outside.

not rare in it. They are not comparable with the earlier sampled figure of 94 in 400 random multilinear pairs ([E14]), which used a different generator.

Provenance. Each witness was computed by two implementations written without sight of each other ([E14]); the refutation of the *inside* form was first recorded as a correction of an earlier belief that it held ([E16], [E17]). The native instrument here is a third implementation, in Rust, from the definitions.

What S5 does not say. Nothing about Proposition 5 of [5], which may hold by another route, and nothing about descended summation-polynomial systems at any n. The note of [E15] argues that Propositions 2 and 3 of [5] exhibit explicit certificates and so do not depend on Lemma 4, and that the inference "fake bound implies true bound" is recoverable per instance when every product in a fake-degree certificate already has unreduced degree at most the fake degree. The justification the paper offers for its Proposition 5, by contrast, runs through Lemma 4 ([E29]); that justification, not the proposition, is what S5 affects. The first-fall-degree assumption itself is a separate matter, on which [4] give evidence against. The result is for characteristic 2 and the equality reading.

7 S6: Assumption 1 of Semaev 2015 off the diagonal

Assumption 1 of [6] asserts $d_{F4} \le 4$ for the Boolean system equivalent to the paper's equation (5) at every $2 \le t \le m$ on the diagonal $k = \lceil n/m \rceil$, where k is the dimension of the subspace $V \subseteq \mathbb{F}_{2^n}$. The programme's goal record states that its experimental basis is 2300 MAGMA systems at $n \le 21$, $m \le 6$ plus one cell at $n = 40$, $m = t = 2$, and that the paper's own Section 4.5.1 reports that the regularity degree "generally exceeds 4 when $k > \lceil n/m \rceil$ though the first fall degree is still 4" ([E21]).

The frozen design of this paper ([E27]) listed "Assumption 1 fails off the diagonal" as a negative result. **We could not establish it.** What the committed records support is three separate facts.

(i) The source's own remark. The off-diagonal behaviour is a remark about the regularity degree in the source paper, not a programme measurement of d_{F4} ([E21]).

(ii) On the diagonal the assumption is not refuted at reachable sizes. With $m = 2$ and V a uniformly random $\mathbb{F}_2$-subspace of dimension exactly $\lceil n/2 \rceil$, the first fall degree is 3.00, flat, for $n = 7, \ldots, 63$, against a shape-matched random Boolean null at 4.00; and a sound refutation predicate for $d_{F4} \le 4$ returns REFUTED at every cell $n = 5, \ldots, 21$, which means the assumption's bound is consistent with every cell reached ([E20]). The same record notes that an earlier degree ladder had measured Frobenius-invariant subspaces of dimension $\mathrm{ord}_n(2)$, which coincide with $\lceil n/2 \rceil$ in 0 of 23 cells, so that ladder says nothing for or against the assumption. First fall and solving degree are not d_{F4}, and the record claims no refutation.

(iii) One off-diagonal closure draw reads $D > 4$. Campaign item 07 measures the degree-4 closure of the chained S_3 descent system with a memory-blocked engine ([E22]). Writing $\delta = l - \lceil n/m \rceil$ for the subspace dimension l of a cell (n, l, m) relative to the diagonal, its measured cells are:

cell (n, l, m)	δ	draws	reading
$(13, 4, 4)$	0	5	$D = 4$ on all
$(11, 4, 4)$	$+1$	4 ($+1$ unreceipted)	$D = 4$ on all
$(13, 5, 4)$	$+1$	2	$D = 4$ on both
$(15, 5, 4)$	$+1$	3 relayed	one draw saturates without 1, $D > 4$; two stop at the time limit, unmeasured
$(19, 6, 3)$	-1	4	$D = 4$ on all

Of the three cells with $\delta = +1$, two read $D = 4$ on every draw; the third, $(15, 5, 4)$ with $N = 50$ unknowns, has one measured draw of four planned that reads $D > 4$. The record itself calls this a one-draw observation, a closure reading rather than a literal F4 reading (no msolve reading exists at that cell), and a result at toy scale; the cell is BLOCKED on time ([E22]).

Proposition 8 (S6, NOT ESTABLISHED)**.** *No committed record contains an F4 degree certificate $d_{F4} > 4$ at any cell, on or off the diagonal. The programme's findings state that Assumption 1 is not refuted (*[E14]*, "reads this before using anything below"), and the only off-diagonal reading above* 4 *is one closure draw in item 07. Items 04 and 05 (*[E36]*,* [E37]*) measure closure degrees of related descent systems at toy cells, the first with the chained S_3 route at $D = 4$ on four cells, the second with grevlex at $D_{\mathrm{mat}} = 4$ on its countable targets; neither is a reading at the deciding cell.*

Why it matters, and what would settle it. The campaign's pricing of binary curves at deployed sizes (item 02) is conditional on constant-degree arms ($D = 3, 4, 6, 7$) that are hypotheses at deployed n, and its gate on item 07 is open ([E26]); the one arm that is the measured growth fit, $D = l + 4$ at $m = 4$, never reaches rho (gap $+105.1$ bits at $n = 131$, rising to $+336.7$ at $n = 571$, Koblitz, weight 1, $K = 1$). Settling S6 needs a degree-5 closure at $(15, 5, 4)$, which the record puts at 2.37×10^6 columns and 3.5×10^{11} bytes of reduced basis, beyond this host, or a different engine. Whether the repository's chained S_3 descent system with a random subspace of dimension l coincides with Semaev's equation (5) up to his conventions is the open definitional question KN-OPEN-d218ec; the sign of δ above is parameter arithmetic, not a statement that the two objects agree.

8 Campaign results that bear on the statements

The blind-spot campaign of 2026-10-18 measured twenty directions. Table 4 lists the ones whose results bear on S1–S6, quoted from each item's own verdict. Items 03, 06, 08, 09, 11, 12, 13, 15 and 16 concern solver, learned-policy, rho-walk, genus-3 linear-algebra and toy-size oracle engineering and bear on none of the six; we cite no number from them. The selection of rows was fixed before the edit (`PROTOCOL.md`, Addendum B), and each row quotes the item's own verdict line and level, as the READMEs read when the ledger digests were taken (item 10's README was still a stage-2 assessment; its frozen rule had no input).

9 Formalisation

Two combinatorial cores are stated in Lean 4 with Mathlib, in a new project `lean/` (beside the register of [E28]) pinned to the same toolchain (`v4.34.1`) and Mathlib revision as the repository's `S3Roots` project. Table 5 is the proof map.

We do not call the new modules machine-checked. The authoring host had neither a Lean toolchain nor the Mathlib cache and too little disk to fetch them, so the proofs were written against the Mathlib interface and have not been elaborated. The repository's proof gate (`tools/lean/proof-gate.sh`) builds the project, fails any theorem whose transitive axioms include `sorryAx`, and accepts only `propext`, `Classical.choice` and `Quot.sound`. The sources contain no `sorry`, `admit`, `native_decide` or user axiom, and open targets are kept in `lean/OPEN_TARGETS.md` rather than in compiled files. Until that gate passes, the Lean statements are specifications of the lemmas, whose mathematical proofs are those given above. The mathematical content is independent of the formalisation; if a Lean proof needs repair the statement stands.

A Lean file is a statement, and its fidelity to the claim must still be read: Lemma 3 is the counting step of Theorem 2, not the theorem, and `sum_yield` is the double count behind Proposition 4, not the multiset instance.

item	verdict as recorded	bears on
01 [E24]	REGRESSES at the constant level; INCONCLUSIVE at the exponent level. No arm cheaper than matched rho at $\log_2 \#E = 24$ to 40; best arms 50.0 [42.0, 60.3] and 51.6 [43.3, 62.0] times matched rho at 40 bits, weight 0	S1 context: distance to rho remains
02 [E26]	ADVANCES (derived, model level), conditional on a constant-degree model; item-07 gate open; not a measured improvement	S6: the constant-degree hypothesis is the unsettled boundary
04 [E36]	ADVANCES, level primitive, stage diagnostic: the Möbius-frame arm lowers the Macaulay closure degree of the incumbent presentation on 5 of 5 cells (for example D from 7 to 5 at $(17, 4, 3)$) and does not go below the chained S_3 arm, $D = 4$ on every cell where it was measured; no S, rho ratio or GAE is computed	S6 context: closure degrees at the cells Assumption 1 concerns
05 [E37]	ADVANCES, level constant, one arm (`wdeg`): 3.4× to 18.6× fewer matrix-entry touches than this engine's grevlex on four cells; the solving degree is not lowered ($D_{\mathrm{mat}} \geq 4$ everywhere) and at the next size the arm hit the time cap	S6 context: grevlex closure degree is 4 on every countable target
07 [E22]	BLOCKED at $(15, 5, 4)$, level primitive; one draw reads $D > 4$	S6 (iii)
10 [E23]	INCONCLUSIVE, level constant; the frozen rule has no input because the held-out, finalist and conservation cards were not run	S3: no evidence from this item
14 [E25]	REGRESSES, level primitive; no $m = 4$ oracle's T_{dec} grows slower than $p^{1/4}$ at the measured scale	S1 context: first measured $m = 4$ cells
17 [E35]	TRADES, level constant: the cyclotomic base is never cheaper than the incumbent by the frozen rule at $n = 83$ (S_{ready} ratio 0.995 to 2.003 across accountings and cells; about 96% of S_{ready} is the arm-independent pair table); model rows put the cyclotomic arm at 231.9 to 1.26×10^7 times the rho floor at $c = 0$ (upper bounds, model rows, not run); the solving-degree leg is BLOCKED	S1 context: distance to rho; S3 context: base-dependent cost is a few percent, yield not measured

Table 4: Campaign items that bear on the six statements. None is a speedup claim.

10 Limitations

- Everything measured is at toy scale; no statement concerns a deployed curve, and none reaches the sizes ($n \geq 131$) at which a speedup would matter.
- S1 is a comparison in probes and table entries against bounds stated in group operations and bits; it is not a proof of genericity, and the structured generic-group bound is vacuous for these rungs.
- S4 has not had an independent review of its proof, and novelty is not adjudicated.
- S5 is for characteristic 2 and the equality reading and says nothing about Proposition 5 of [5].
- S6 is open. We include it so that a reader does not count it among the closed negatives.
- Several evidence files were produced by sibling repositories whose heads moved during the work; Appendix A fixes the digests read.
- The new Lean modules have not been kernel-checked (Section 9).

statement	Lean declaration	status
Lemma 3(i)	`IcNegative.agree_card_le_one`	written, **not yet kernel-checked**
Lemma 3(ii)	`badSecrets_card_le`, `badSecrets_card_le_sq`	written, not yet kernel-checked
Lemma 3(iii)	`injOn_of_not_bad`	written, not yet kernel-checked
Prop. 4, any map	`sum_yield`, `sum_yield_cast`, `total_yield_eq`	written, not yet kernel-checked
coverage bound	`coverage_le_card_dom`, `coverage_le_card_cod`	written, not yet kernel-checked
Prop. 4, list form	`FactorBaseDesigns.sum_count`, `patterns_neutral`	kernel-checked in CI [E7], [E8]
Theorem 2 in full	—	open (O1)
Sym^m binomial count	—	open (O2)
Prop. 7	—	open (O3); finite check is native
Theorem 5	—	open (O4)

Table 5: Proof map. "Not yet kernel-checked" means the proofs were written on a host with no Lean toolchain and no Mathlib cache; the repository proof gate is the first check.

A Evidence ledger

Each row is a file read for this paper, with the first 16 hex digits of its SHA-256. The full digests are in `evidence/ledger.full`; the ledger is regenerated by `evidence/make_ledger.sh`. Repository heads when generated are in `EVIDENCE.md`.

id	repository:path	sha256 (16)
E1	`crypto:research/lit2026_w3_ggm_audit_20261011/reports/run-20261011T2012Z/table.json`	`26d462fe6f6de45c`
E2	`crypto:research/lit2026_w3_ggm_audit_20261011/reports/run-20261011T2012Z/report.md`	`2062646518f95046`
E3	`crypto:research/lit2026_w3_ggm_audit_20261011/README.md`	`4a023d24168ddda6`
E4	`crypto:RESEARCH_ISOGENY_FIBERS_ECDLP.md`	`23205b00c7cc4c25`
E5	`crypto-autoresearcher:knowledge/findings/KN-FIND-007.md`	`9dd306d2fad7387d`
E6	`crypto:research/cyclotomic_falsifier_20261015/README.md`	`be5199e1a20534e0`
E7	`crypto:research/factor_base_designs_20261015/lean/FactorBaseDesigns/Yield.lean`	`e325cb42e2c560fc`
E8	`crypto:research/factor_base_designs_20261015/proof_logs/axioms.txt`	`06148bb5c423bf9f`
E9	`crypto-autoresearcher:knowledge/findings/KN-FIND-a8990a.md`	`aaefb551e0fbee1c`
E10	`crypto-autoresearcher:papers/semaev-conservation-specialization/paper.tex`	`6ce58081151c6e06`
E11	`crypto-autoresearcher:experiments/EXP-SMON-e5cbe6/runs/RUN-SMON-e5cbe6-001/raw-result.json`	`a7f3cb1265cf1102`
E12	`crypto-autoresearcher:ledger/corrections/CORR-20260807-652652.yaml`	`67da07f933b70565`
E13	`crypto-autoresearcher:knowledge/findings/KN-FIND-6edecd.md`	`935ede09e62a5abf`
E14	`crypto-autoresearcher:knowledge/findings/KN-FIND-936151.md`	`a17623ab3189cba0`
E15	`crypto-autoresearcher:papers/nagao-lemma4-note/note.md`	`300372bab1ebd7fc`
E16	`crypto-autoresearcher:ledger/corrections/CORR-20260921-942a62.yaml`	`f9def986c5ec51cc`
E17	`crypto-autoresearcher:tools/lemma4_inside_form.py`	`779e865265a768a7`
E18	`crypto:papers/ic-negative-results/evidence/checks.txt`	`ddd155bb68e11e12`
E19	`crypto:examples/blindspots_19_checks.rs`	`751277a8c075483c`
E20	`crypto-autoresearcher:knowledge/findings/KN-FIND-9cba50.md`	`812688a01a31d18a`
E21	`crypto-autoresearcher:ledger/goals/GOAL-SEMBIN-5078bc/goal.yaml`	`107a5cc6507e58ff`
E22	`crypto:research/blindspots_20261018/07-blocked-f4-degree-cells/README.md`	`131ad266f6cb604b`
E23	`crypto:research/blindspots_20261018/10-kernel-program-search/README.md`	`67dd70829c4e08a7`
E24	`crypto:research/blindspots_20261018/01-fp2-gaudry-k2/README.md`	`34d06d7bdec87acc`
E25	`crypto:research/blindspots_20261018/14-prime-m4-cells/README.md`	`5a333ca46a54939f`
E26	`crypto:research/blindspots_20261018/02-deployed-binary-pricing/README.md`	`529c38f03889ab2b`
E27	`crypto:research/blindspots_20261018/EXPERIMENTS.md`	`fe91cf9a1abd32fb`
E28	`crypto:research/factor_base_designs_20261015/README.md`	`7aec9d11222cf315`
E29	`crypto-autoresearcher:coordination/review/sembin-20260916-propquant/report.md`	`a0c7642720eef546`

id	repository:path	sha256 (16)
E30	crypto:research/blindspots_20261018/11-learned-prefilter/README.md	943422dff35ed15f
E31	crypto:papers/ic-negative-results/lean/IcNegative/AffineCore.lean	178ea2dd12090590
E32	crypto:papers/ic-negative-results/lean/IcNegative/YieldConservation.lean	ee973d4cd005356f
E33	crypto:research/pair46_factor_bases_20261015/13-xy-windows/RESULTS.md	f2f6a266dc807687
E34	crypto-autoresearcher:papers/semaev-conservation-specialization/README.md	99888c7d58c128e6
E35	crypto:research/blindspots_20261018/17-cyclotomic-cost-n83/README.md	7ba9a2653dce7a3d
E36	crypto:research/blindspots_20261018/04-presentation-search/README.md	d4676ae4b5ec3e4b
E37	crypto:research/blindspots_20261018/05-elimination-orders/README.md	845129e7617bb62d

B Reproducing the native checks

```
rustc --edition 2021 -C opt-level=3 -o b19 examples/blindspots_19_checks.rs
prlimit --as=4294967296 ./b19 all > output.txt   # about 0.3 s, one thread
diff <(tail -n +3 output.txt) \
     <(tail -n +3 papers/ic-negative-results/evidence/checks.txt)
```

The output ends with `ALL CHECKS PASS`. The subcommands are `lemma4`, `affine` and `yield`.

References

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[4] M.-D. A. Huang, M. Kosters and S. L. Yeo. Last fall degree, HFE, and Weil descent attacks on ECDLP. CRYPTO 2015 (ePrint 2015/573).

[5] K. Nagao. Complexity of ECDLP under the first fall degree assumption. ePrint 2015/984.

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